

n-doping of organic semiconductors catalysed by organometallic complexes

Received: 12 April 2025

Accepted: 27 October 2025

Published online: 12 December 2025



Sergio Gámez-Valenzuela¹, Jianfeng Li¹, Kui Feng¹, Bin Liu¹, Suxiang Ma¹, Yongchun Li¹, Yani Lu¹, Xiage Zhang¹, Bolin Li¹, Sang Young Jeong², Hong Wang¹, Han Young Woo², Hanqiang Wang^{1,3}✉, Alessandro Motta⁴✉ & Xugang Guo¹✉

Molecular doping is essential for optimizing carrier concentration, charge mobility, and energy levels of organic semiconductors (OSCs), thereby enhancing device performance. However, efficient n(electron)-type doping remains challenging, as conventional techniques often fail to achieve high doping efficiency with minimal counterion-induced disorder under mild conditions. Here, we present a catalysed n-doping strategy for OSCs using air-stable, cost-effective, and commercially available soluble organometallic complexes, such as Pt(COD)Cl₂, which enable rapid n-dopant activation with just 10 s of annealing at 120 °C. This approach demonstrates excellent generality across diverse OSCs, dopants and catalysts, achieving electrical conductivities exceeding 230 S cm⁻¹ with up to 10-fold improvements over existing catalyst-assisted n-doping methods. Importantly, it mitigates counterion-induced structural disorder in doped OSCs, leading to a high power factor of 175 μW m⁻¹ K⁻² and thermoelectric figure of merit (ZT) of 0.43 at room temperature for catalytically *N*-DMBI-doped polymer films. This strategy addresses long-standing challenges in n-doping while unlocking new avenues for next-generation organic electronics.

Molecular doping has become one of the most common and efficient approaches for optimizing charge carrier concentration and reducing charge injection/extraction barriers in organic electronic devices, including thin-film transistors^{1,2}, light-emitting diodes^{3,4}, solar cells^{5,6}, thermoelectrics^{7,8}, and electrochemical devices^{9,10}. However, despite considerable efforts made in both p(hole)-type and n(electron)-type doping processes, the latter still remains far more challenging particularly due to the limited stability and low doping efficiency of n-dopants (often < 10%)¹¹. To address these issues, a new approach has been devised using stable precursor-type molecular dopants (see Fig. 1a, for the example of *N*-DMBI-H dopant). Once incorporated into the host semiconductor matrix, these dopants undergo post-deposition thermal- or photo-activation to generate in-situ active anionic and/or radical species capable of n-doping organic

semiconductors (OSCs)^{12,13}. Nevertheless, the associated high activation energy typically required for in-situ generation of active doping species represents a major barrier to extending the scope of this strategy. For example, the widely used benzoimidazole-based molecular dopant *N*-DMBI-H generally requires prolonged annealing times (*t*_{ann}) of over 2 h at temperatures (*T*_{ann}) between 120 and 150 °C to enable effective n-doping^{14,15}. Such conditions impose great obstacles for commercial applications of OSCs and their implementation on flexible plastic substrates.

To tackle this limitation, some of us pioneered catalysed n-doping methodology, demonstrating that vapor-deposited transition metal nanoparticles (NPs) can greatly promote the heterolytic C-H bond cleavage of hydride-based n-dopant precursors, thereby enhancing n-doping efficiency within much shorter timescales¹⁶. Despite

¹Department of Materials Science and Engineering, Southern University of Science and Technology (SUSTech), Shenzhen, Guangdong, China. ²Department of Chemistry, Korea University, Seoul, Republic of Korea. ³Department of Chemistry and Dongguan Key Laboratory for Data Science and Intelligent Medicine, Great Bay University, Dongguan, Guangdong, China. ⁴Dipartimento di Chimica, Università di Roma “La Sapienza” and INSTM, UdR Roma, Rome, Italy.

✉ e-mail: hanqiangwang@gbu.edu.cn; alessandro.motta@uniroma1.it; guoxg@sustech.edu.cn

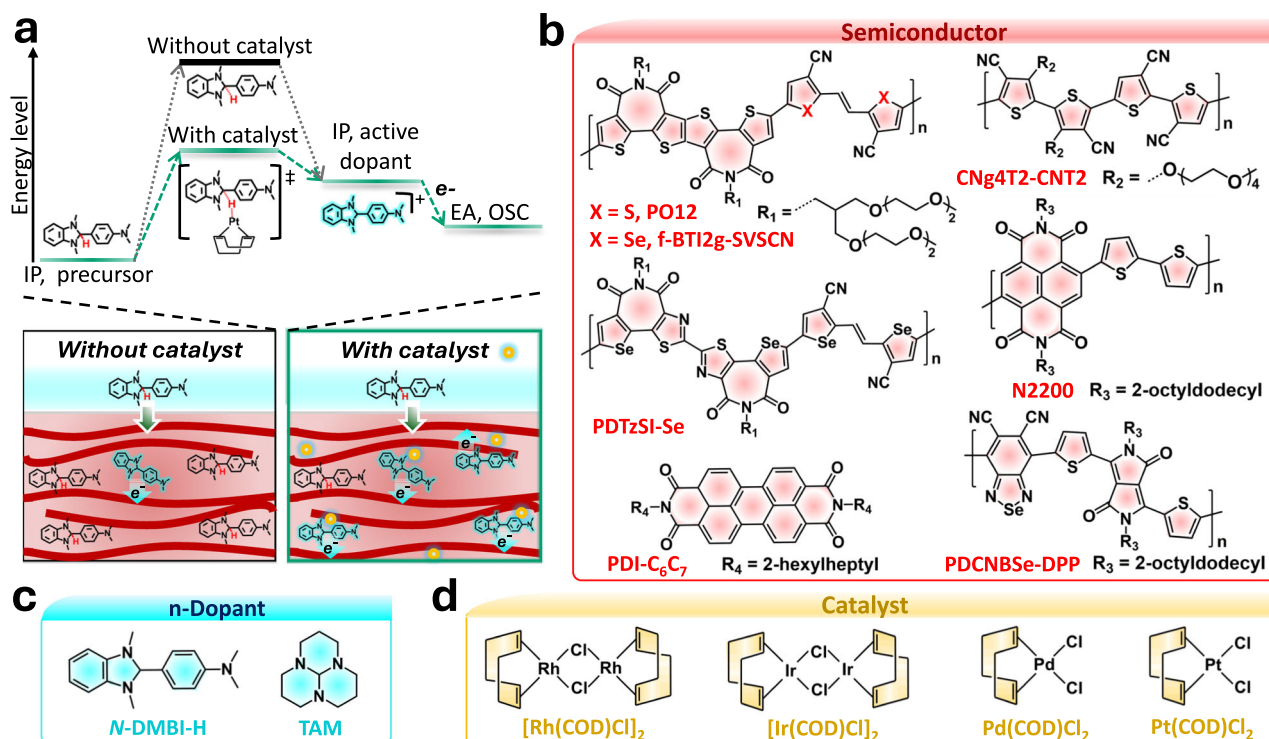


Fig. 1 | Concept of solution-processable organometallic-catalysed n-doping. **a** Schematic representation of the catalytic doping process for hydride-based n-dopants. In their initial state, the deep ionization potential (IP) of precursor-type dopants thermodynamically restricts direct electron transfer to organic semiconductors with low electron affinity (EA), necessitating C–H bond cleavage to generate in-situ active doping species before electron transfer takes place. However, the high activation energy required for heterolytic bond cleavage limits the

doping reaction rate and overall doping efficiency. Incorporation of solution-processable organometallic catalysts significantly lowers the activation energy for bond cleavage, thereby increasing the doping reaction rate and enhancing the doping efficiency. **b–d** Chemical structures of the n-type organic semiconductors (**b**), hydride-based molecular dopants (**c**), and organometallic catalysts (**d**) used in this study.

improving n-doping efficiency, this approach suffers from several key drawbacks: relying on costly vapor deposition, unsuitable for printed electronics; confining the catalytic process to the area near the surface of pre-deposited nanoparticles, which limits the thickness of catalytically doped films; using highly active catalysts like gold nanoparticles (AuNPs) that complicate doping control and device integration. In an attempt to overcome these limitations and advance toward practical applications, Facchetti *et al.* recently explored a new type of AuNPs functionalized on the surface with alkylthiol solubilizing groups (F-AuNPs) as catalysts¹⁷. This method provides access to soluble catalysts that remain inactive until some of the surface ligands are removed through mild thermal activation. However, this approach not only yields lower n-doping efficiencies compared to its predecessor but also relies on the meticulous functionalization of pre-evaporated NPs, limiting its general applicability. Concurrently, Fabiano *et al.* demonstrated photocatalytic n-doping of the prototypical polymer poly(benzimidazobenzophenanthroline) (BBL) using acridinium-based photocatalysts (PCs). In the presence of light, these PCs catalyze the activation of the weak reductant Et₃N, achieving electrical conductivities (σ) up to 1 S cm^{−1}¹⁸. However, this approach is primarily developed for p-doping, as only the BBL polymer with a low-lying lowest unoccupied molecular orbital (LUMO) energy level of −4.3 eV was shown to be n-doped by Et₃N, resulting in a σ value 8 times lower than that reported for uncatalysed n-doping of the same polymer¹⁹. Therefore, the challenge remains in developing a universally applicable solution-processable strategy to effectively n-dope OSCs.

In this study, we fill this vast gap by developing a simple and general n-doping strategy based on solution-processable catalysts. Our approach demonstrates that incorporating 1,5-cyclooctadiene (COD)-based organometallic complexes such as Pt(COD)Cl₂

significantly reduces the activation energy of precursor-type dopants like N-DMBI-H, resulting in dramatic improvements in doping efficiency and σ . These catalysts were selected for their combination of excellent solubility, air stability, commercial availability, and relatively low cost (e.g., ~\$206 per 1 × g for Pt(COD)Cl₂)^{20,21}. The COD ligand plays a key role, as it ensures good solubility in commonly used organic solvents without significantly increasing the molecular size. This compact molecular size may facilitate efficient diffusion and uniform dispersion within the semiconductor matrix, helping to minimize microstructural disruption of polymer films during processing²². To validate this strategy, we conducted catalytic doping studies over a diverse range of n-type OSCs (Fig. 1b) and molecular dopants (Fig. 1c) using four different COD-derivatives (Fig. 1d), dichloro(1,5-cyclooctadiene)platinum(II) (Pt(COD)Cl₂), dichloro(1,5-cyclooctadiene)palladium(II) (Pd(COD)Cl₂), chloro(1,5-cyclooctadiene)rhodium(I) dimer [Rh(COD)Cl]₂ and chloro(1,5-cyclooctadiene)iridium(I) dimer [Ir(COD)Cl]₂. Their catalytic performance was superior to the state-of-the-art (photo)catalytic n-doping methods, establishing our approach as a new paradigm for the n-doping of OSCs and the fabrication of high-performance organic electronic devices.

Results and discussion

Pt(COD)Cl₂ catalysed n-doping of PO12

The solution-processable catalysed n-doping approach is straightforward and highly efficient, as demonstrated in Fig. 2 for the case-study Pt(COD)Cl₂-catalysed doping of PO12 by N-DMBI-H. Unless otherwise indicated, the doping of films was carried out following a Seq-D doping protocol, where a blend solution of semiconductor and catalyst, with a specific catalyst molar ratio (0.5–10 mol%), was first spin-coated, followed by the deposition of a 5 mg mL^{−1} N-DMBI-H dopant solution onto

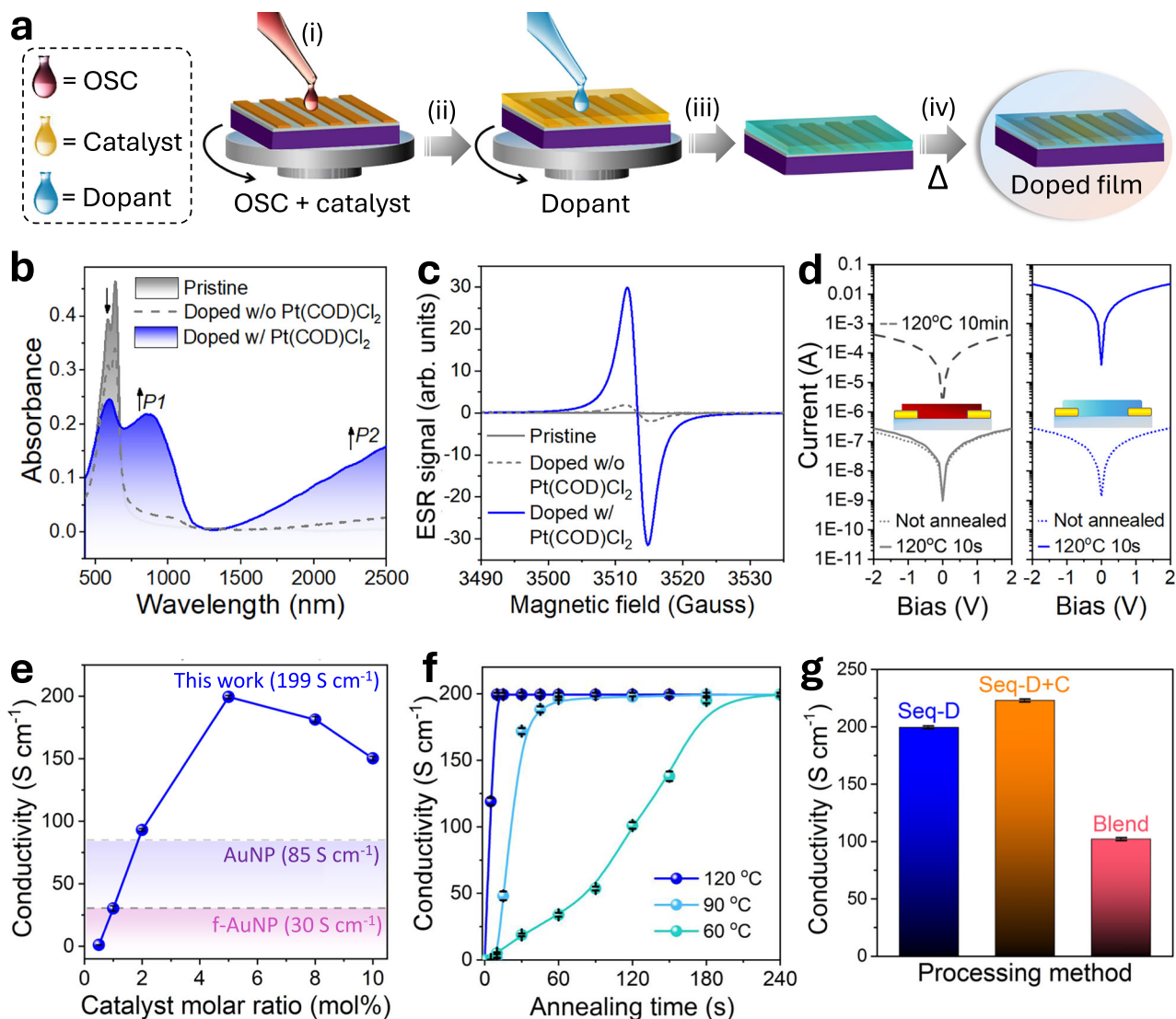


Fig. 2 | Pt(COD)Cl₂ catalysed n-doping of PO12 with N-DMBI-H. **a** Schematic illustration of the catalytic n-doping method based on organometallic complexes: (i) Mixing the catalyst with the semiconductor at a catalyst/semiconductor molar ratio of 0.5–10 mol%; (ii) Spin-coating the resulting solution; (iii) Depositing a 5 mg mL⁻¹ n-dopant solution onto the film; (iv) Doping activation by thermal annealing at 120 °C for 10 s. **b** UV-vis-NIR absorption spectra and **c** ESR signals of undoped PO12 films, N-DMBI-H doped PO12 films without catalyst, and with 5 mol% Pt(COD)Cl₂. ESR measurements were performed at an X-band frequency of 9.84 GHz and at room temperature. **d** Representative current-voltage curves of N-DMBI-H doped PO12 films without (gray lines) or with 5 mol% Pt(COD)Cl₂ (blue lines). Dotted lines represent measurements before thermal activation, solid lines after 10 s of thermal activation, and the dashed gray line after 10 min of thermal annealing. These results demonstrate that catalytic doping activates only after thermal annealing at 120 °C for a short time of 10 s. Even after an annealing time of

10 min at 120 °C, the uncatalysed doping remained inefficient, as evidenced by the low σ values. **e** Electrical conductivity of sequentially n-doped PO12 films as a function of the Pt(COD)Cl₂ catalyst molar ratio. The maximum σ values reported for the same polymer using AuNPs and f-AuNPs catalysts are indicated by horizontal lines for comparison. **f** Temporal evolution of the conductivity of N-DMBI-H doped PO12 films with 5 mol% Pt(COD)Cl₂ catalyst when annealed at different temperatures. **g** Comparison of the highest electrical conductivities achieved for N-DMBI-H doped PO12 films using different doping protocols: Seq-D, Seq-D + C (spin-coating of the semiconductor solution followed by deposition of a dopant + catalyst blend solution), and blend doping (directly spin-coating a ternary blend solution containing the OSC, catalyst, and dopant). Unless otherwise indicated, doping activation was performed by annealing at 120 °C for 10 s (with catalyst) or 10 min (without catalyst). Error bars for conductivity indicate the standard deviation of five independent samples.

the film (Fig. 2a). Control experiments were conducted by doping the semiconductor in the absence of catalyst, with all doping processes activated by mild thermal annealing at 120 °C for 10 s (the detailed doping procedure is described in the Experimental “Methods” section). We first discuss the catalysed n-doping reaction of PO12 as a proof-of-concept of our new catalytic doping strategy. Despite its acceptor-acceptor type backbone and relatively deep LUMO energy level of -3.81 eV²³, polymer PO12 can only be weakly n-doped even at a high N-DMBI-H loading of 5 mg mL⁻¹ and annealing at 120 °C for 10 min. This poor doping efficiency was reflected by an almost negligible polaron signal in the UV-Vis-NIR absorption spectrum (Fig. 2b) and a

low charge carrier density of $\sim 8 \times 10^{17}$ cm⁻³ (Fig. 2c) determined by electron spin resonance (ESR). As a result, the σ only reached 2.9 ± 0.2 S cm⁻¹ after thermal activation for 10 min (Fig. 2d). However, the presence of Pt(COD)Cl₂ enabled efficient n-doping of the PO12 film with N-DMBI-H after annealing at 120 °C for only 10 s. This was evidenced by the bleaching of the 500–700 nm charge transfer/ π - π^* neutral absorption peaks, the emergence of strong polaronic ($p1$ and $p2$) near-infrared absorption bands (Fig. 2b), and a remarkably high ESR signal (Fig. 2c). Furthermore, catalysed n-doping of PO12 resulted in a three orders of magnitude enhancement in charge carrier density, reaching $\sim 1 \times 10^{21}$ cm⁻³, an underestimated value if bipolarons are present

(Supplementary Fig. S1)²⁴. Consequently, at the short annealing time of 10 s, the σ of catalytically doped PO12 was substantially improved by more than three orders of magnitude, rising from $10^{-3} \text{ S cm}^{-1}$ for uncatalysed doping to over 1 S cm^{-1} at the minimal Pt(COD)Cl₂ loading of 0.5 mol%. Notably, further increasing the catalyst loading to 5 mol% resulted in a record σ of $199.4 \pm 1.3 \text{ S cm}^{-1}$ (Fig. 2d, e), clearly exceeding the value of 30 and 85 S cm^{-1} reported by the f-AuNPs- and AuNPs-catalytic n-doping methods^{17,23}. It is noteworthy that the σ value achieved using this new approach stands among the highest reported for n-doped OSCs. More importantly, a direct comparison of doped films with (Fig. 2d, blue dashed line) and without (Fig. 2d, grey dashed line) the Pt(COD)Cl₂ catalyst reveals no discernible change in σ prior to thermal activation, underscoring that this approach enables a fully controllable catalytic n-doping process that becomes active exclusively after thermal annealing (Fig. 2d, blue solid line). This method overcomes a key limitation of the state-of-the-art AuNPs-based doping strategy, whose catalytic activity is too high and cannot be controlled.

We also noticed that this approach allows for precise control of doping kinetics by modulating the annealing temperature, which can be reduced to 90 °C or even 60 °C with reasonably short annealing times of 1 and 3 min, respectively (Fig. 2f). This advancement facilitates efficient n-doping while bypassing the need for high-temperature annealing associated with conventional doping strategies, which typically exhibit a dramatic performance drop at temperatures below 80 °C¹⁴. Therefore, this new method not only enhances doping efficiency but also paves the way for integrating the doping strategy into standard fabrication processes.

Since a sequential doping protocol requires a separate deposition of the organic semiconductor and molecular dopant, the way of introducing the catalyst into thin films becomes a critical and intriguing question to address. We first explored this effect using the Seq-D + C doping method in which the semiconductor film is first spin-coated, followed by the deposition of a dopant + catalyst blend solution. To our delight, this method resulted in a further enhancement in σ , achieving an impressive value of $222.9 \pm 1.1 \text{ S cm}^{-1}$ at a catalyst loading of 5 mol% (Fig. 2g). However, a σ of $7.8 \pm 1.1 \text{ S cm}^{-1}$ was observed even without deliberate n-doping activation via thermal annealing, suggesting the in-situ generation of active catalytic species as soon as both catalyst and dopant solutions become in contact, as discussed in the next sections (Supplementary Fig. S2, 3). Additionally, the blend doping method, in which a semiconductor + dopant + catalyst ternary blend solution is directly spin-coated, was also tested. As expected, this approach yielded significantly lower σ values compared to the sequentially doped films (Supplementary Table S1), likely due to the disruptive effect on film microstructure induced by dopant molecules using this method²⁵. However, the blend doping strategy offers a key intrinsic advantage: it allows for a more precise determination of n-doping efficiency, as the molar ratio of molecular dopant to polymer repeat units can be accurately defined in the precursor solution. Accordingly, blend doped PO12 films showed a high doping efficiency of up to 69.5% at the optimal Pt(COD)Cl₂ loading of 5 mol%, which also yielded the highest σ of $103.2 \pm 1.4 \text{ S cm}^{-1}$ (Supplementary Fig. S4). This efficiency surpasses that achieved with previously reported AuNP-catalysed n-doping strategies (63.3%)²³. Based on these findings, the Seq-D protocol was selected for subsequent experiments, as it provides enhanced control over the doping process while minimizing the adverse effects of blend doping on film morphology and microstructure.

Generality of the organometallic-catalysed n-doping

To demonstrate the generality of our strategy, we applied the Pt(COD)Cl₂ catalysed n-doping method to a wide range of OSCs featuring diverse conjugated backbones functionalized with either hydrophilic or hydrophobic side chains. These included selenophenylthiazole

imide-based polymer PDTzSI-Se²⁶, cyano-functionalized polythiophene derivative CNg4T2-CNT2²⁷, diketopyrrolopyrrole (DPP)-based polymer PDCNBSe-DPP²⁸, and well-established naphthalene diimide (NDI) or perylene diimide (PDI)-based materials such as the benchmark polymer N2200 and small molecule PDI-C₆C₇ (Fig. 1b and Supplementary Fig. S5). In this context, the broad applicability of our method to OSCs was clearly confirmed through absorption and electrical conductivity measurements (Fig. 3). In particular, the emergence of pronounced polaronic transitions in polymers and radical-anion absorption in the small molecule PDI-C₆C₇ corroborates that all tested semiconductors are effectively doped by *N*-DMBI-H after just 10 s annealing (Fig. 3a). This rapid doping process is evidenced by a remarkable color change in the films, confirming efficient permeation of the doping reaction throughout the entire film thickness (Supplementary Table S2 and Fig. S6). For PDTzSI-Se, catalytically doped films led to a substantial improvement in σ (Fig. 3b), achieving up to a 17-fold increase over the uncatalysed *N*-DMBI-H doped films (rising from $12.9 \pm 1.9 \text{ S cm}^{-1}$ to an impressive record σ of 217.9 S cm^{-1}). Notably, this value is among the highest reported for *N*-DMBI-H doped OSCs²⁵. Our catalytic doping approach proves highly effective even for intrinsically difficult-to-dope polymers such as PDCNBSe-DPP and N2200, which achieved 67- and 100-fold σ enhancements compared to uncatalysed *N*-DMBI-H doped films, respectively (Supplementary Table S3 and Supplementary Fig. S7, 8). More importantly, all the catalytically doped samples in this work show 2 to 4 times higher σ values than those reported using the current leading catalytic n-doping techniques (Supplementary Table S4)^{16,17}. Particularly remarkable is the case of the benchmark polymer N2200, where conventional catalytic methods fail to increase σ , yielding values that are 1–2 orders of magnitude lower than those achieved with our approach¹⁷.

We also expanded the applicability and demonstrated that our new n-doping method is not exclusive to the Pt(COD)Cl₂ catalyst. Other COD-based organometallic complexes containing group 9/10 transition metals, such as Pd(COD)Cl₂, [Rh(COD)Cl]₂, and [Ir(COD)Cl]₂, can also effectively catalyse the doping process (Fig. 3c, d). The catalytic activity of these complexes was assessed using UV-vis-NIR absorption spectra, where a more efficient catalyst is expected to produce stronger polaron or radical-anion absorption while weakening the charge-transfer/ π - π^* absorption bands of the neutral state. Accordingly, catalytically doped films demonstrate significant variations in polaron/ π - π^* intensity ratios depending on the catalyst used (Fig. 3c and Supplementary Fig. S9), qualitatively establishing the order of catalytic efficiency as Pt(COD)Cl₂ > [Rh(COD)Cl]₂ > [Ir(COD)Cl]₂ > Pd(COD)Cl₂. This trend is further reflected in the electrical conductivity of the resulting n-doped films, which varies markedly with both catalyst identity and loading (Fig. 3d and Supplementary Table S5). At low concentrations (< 2 mol%), [Rh(COD)Cl]₂ and [Ir(COD)Cl]₂ yield higher conductivities than Pt(COD)Cl₂, likely due to the dimeric nature of the Rh(I) and Ir(I) complexes, which increases the effective concentration of active species in solution. However, this trend reverses at higher loadings (> 5 mol%), where Pt(COD)Cl₂ yields the highest σ , followed by [Rh(COD)Cl]₂, [Ir(COD)Cl]₂, and Pd(COD)Cl₂. The decline in performance of the Rh(I) and Ir(I) complexes at higher concentrations may result from catalyst aggregation or polymer microstructural disruption, adversely affecting charge transport^{29,30}. In contrast, the monomeric Pt(COD)Cl₂ maintains high activity over a broader concentration range. Its superior performance aligns with its well-documented ligand exchange lability and high propensity for oxidative addition reactions²¹, both of which are key to enabling efficient heterolytic activation of the *N*-DMBI-H dopant. By comparison, Pd(COD)Cl₂ is known to undergo ligand exchange with acetonitrile (the solvent used) to form bis(acetonitrile)palladium(II) dichloride, which may reduce the concentration of catalytically active species in solution and account for its comparatively low

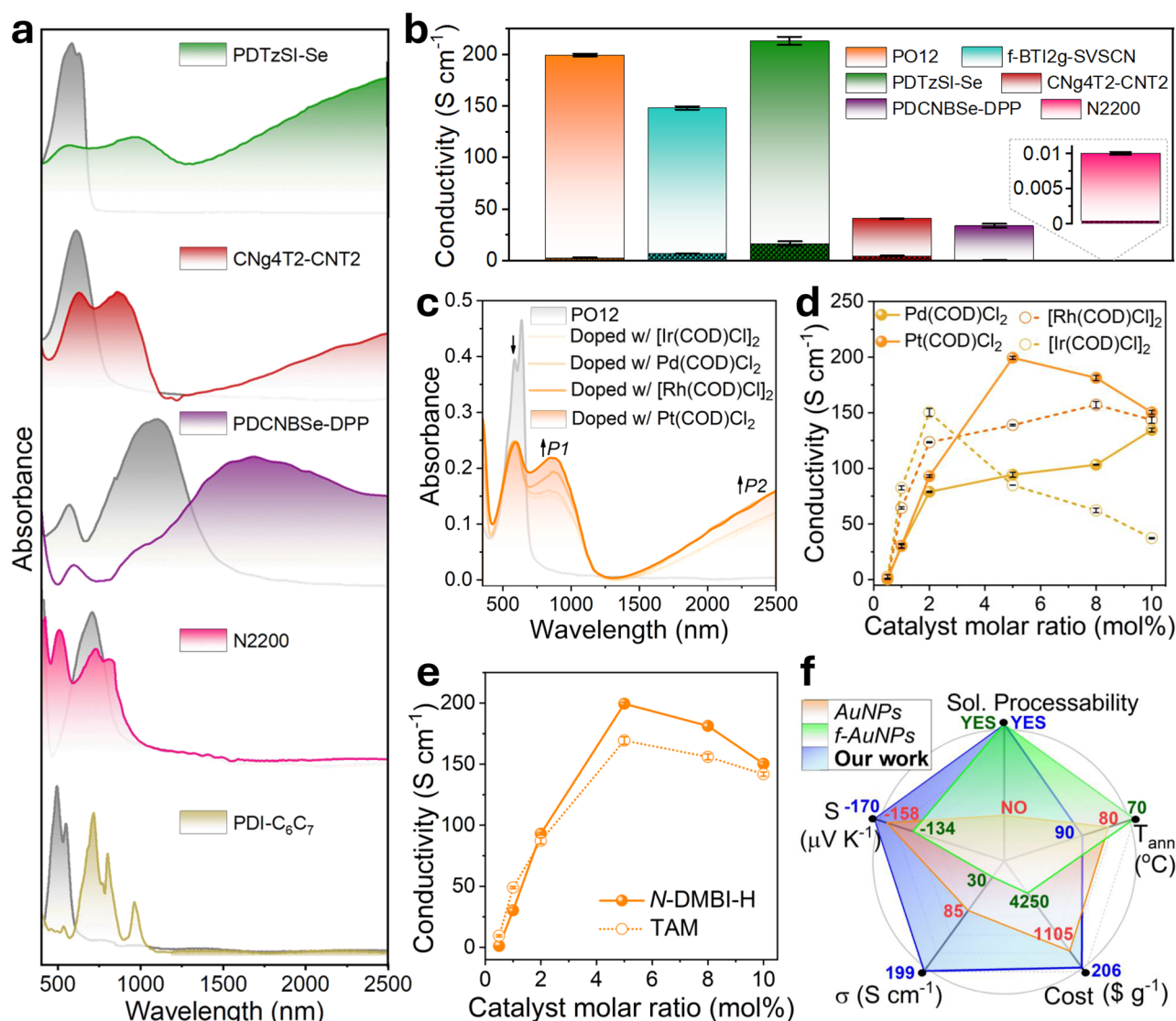


Fig. 3 | Generality of the organometallics-catalysed n-doping method.

a UV-vis-NIR absorption spectra of pristine (grey lines) and catalytically *N*-DMBI-H doped films of various OSCs, including PDTzSI-Se (green line), CNg4T2-CNT2 (red line), PDCNBSe-DPP (purple line), N2200 (pink line), and PDI-C₆C₇ (brown line) (see Supplementary Table S2 for details). **b** Comparison of the σ values measured for uncatalysed *N*-DMBI-H doped polymer films (dark pattern) and those doped using Pt(COD)Cl₂ as catalyst (light pattern). **c** UV-vis-NIR absorption spectra and **d** electrical conductivity of PO12 films after catalytic *N*-DMBI-H doping using different COD-based organometallic complexes, such as those featuring transition metals rhodium ([Rh(COD)Cl]₂), iridium ([Ir(COD)Cl]₂), or palladium (Pd(COD)Cl₂). **e** Comparison between the average electrical conductivities of sequentially *N*-DMBI-H- (solid line) and TAM-doped (dash line) PO12 films using Pt(COD)Cl₂ as

catalyst. **f** Radar chart for the comprehensive comparison of our new n-doping strategy versus previously reported catalytically assisted n-doping methods. The chart includes: (i) the lowest thermal activation temperatures required for effective doping within a practical time of 1 min, (ii) catalyst cost based on current commercial sources, (iii) the highest σ values achieved for n-doped PO12 films, (iv) Seebeck coefficient at a comparable σ of 30 S cm⁻¹, and (v) suitability for solution processability (see Supplementary Fig. S12 and Table S7 for detailed discussion). Unless otherwise indicated, doping activation was performed by annealing at 120 °C for 10 s (with catalyst) or 10 min (without catalyst). A dopant concentration of 5 mg mL⁻¹ was used for all cases. Error bars for conductivity indicate the standard deviation of five independent samples.

conductivities^{31,32}. These findings highlight the importance of selecting a catalyst not only based on intrinsic electronic and kinetic properties but also considering its compatibility with the processing conditions.

Intriguingly, it was found that all the OSCs investigated in this study can be catalytically doped by other precursor-type n-dopants, such as the triaminomethane-based TAM (Fig. 3e and Supplementary Fig. S10, 11)¹³, demonstrating comparable σ values with either of the two molecular dopants used (Supplementary Table S6). These results highlight the broad applicability and competitive performance of our catalytic n-doping strategy in comparison to previously reported approaches (Fig. 3f and Supplementary Table S7).

Mechanism of catalysed n-doping

The unique characteristics of our doping approach prompted us to unravel the hydride/electron transfer processes that drive the thermodynamically facilitated C-H bond cleavage reaction, through a battery of spectroscopies and theoretical modeling. For this purpose, we used Pt(COD)Cl₂ and *N*-DMBI-H as catalyst and n-dopant, respectively. The small molecule PDI-C₆C₇ was chosen as a representative n-type semiconductor due to its excellent suitability for spectroscopic monitoring, managing computation time, and direct comparison with existing literature^{16,17}. Close analysis of ¹H nuclear magnetic resonance (¹H-NMR) spectra revealed that the solubilizing cyclooctadiene group

remains coordinated to the Pt metal center in solution, even after heating above 60 °C for 24 h in the presence of PDI-C₆C₇. This coordination preserves the complex's integrity and solubility for over a week, contrasting sharply with the property of its non-COD analogue, PtCl₂, which is completely insoluble in most organic solvents (Supplementary Fig. S13). However, the introduction of the *N*-DMBI-H dopant leads to cleavage of the Pt–Cl bond in Pt(COD)Cl₂ due to hydride transfer induced Pt–H bond formation. This is evidenced by the absence of $\nu(\text{Pt-Cl})$ stretching modes in the Raman spectrum of the ex-situ reaction product of Pt(COD)Cl₂ with *N*-DMBI-H (Supplementary Fig. S14) and the appearance of two new Cl 2*p* peaks derived from the generation of ionic chloride (Supplementary Fig. S15). This observation is consistent with previously reported phenomena in Pt-based films during H₂ detection, where hydrides induce Pt reduction and Pt–H bond formation^{33–35}. Further validation comes from the isolation and characterization of the reaction product between Pt(COD)Cl₂ and *N*-DMBI-H. X-ray diffraction of single crystal structure and ¹H-NMR analyses confirmed the formation of *N*-DMBI⁺Cl[−] (Supplementary Fig. S16, 17), indirectly pointing to the generation of Pt(COD)H₂. Therefore, chloride ligands are proposed to act as pre-activation elements in the hydride transfer process, enabling the in-situ generation of Pt(COD)H₂ as the active catalyst in the following n-doping reaction. Importantly, because of their strong ionic interaction with the cationic dopant and the intrinsically low ionic conductivity of Cl[−] in solid-state polymer matrices, these anions are unlikely to make any measurable contribution to the electrical transport of the doped films, which is instead dominated by electronic carriers (see Supporting Information, page S22, for further discussion). Supporting this mechanistic proposal, density functional theory (DFT) calculations confirm that the reaction between Pt(COD)Cl₂ and *N*-DMBI-H is exergonic, with a calculated ΔG of $-7.1 \text{ kcal mol}^{-1}$ (see Supplementary Fig. S22). Additionally, in-situ electrospray ionization mass spectrometry (ESI-MS) of the solution reaction revealed a cation at m/z 266.1652 (*N*-DMBI⁺) and anions at m/z 304.0671 ([Pt(COD)H][−]) and 306.0827 ([Pt(COD)H₃][−]) (Fig. 4a and Supplementary Fig. S18). These findings collectively corroborate that the reaction mechanism operates through the formation of active hydride-based platinum complexes.

In light of our experimental findings and previous studies on the *N*-DMBI-H doping reaction^{16,17,36–38}, a plausible reaction mechanism for the Pt(COD)Cl₂-catalysed *N*-DMBI-H doping of PDI-C₆C₇ is proposed (Fig. 4d). The catalytic cycle is assumed to initiate with Pt(COD)H₂, which facilitates a hydride transfer from the dopant resulting in the formation of the ion-pair adduct (*N*-DMBI⁺/[Pt(COD)H₃][−]). Subsequently, [Pt(COD)H₃][−] transfers an electron to the PDI-C₆C₇ semiconductor, generating anion radical species PDI-C₆C₇^{•−}, the n-doped product. This step aligns with the prominent radical anion signals observed in the UV–Vis–NIR spectra of doped PDI-C₆C₇ films (Fig. 4b). The resulting intermediate, [Pt(COD)H₃][−], undergoes spontaneous reductive elimination of H₂, leading to the formation of [Pt(COD)H][−]. The formation of H₂ as a key reaction product is confirmed by ¹H-NMR analysis (Fig. 4c and Supplementary Fig. S19). At this point, the first two steps are repeated: (i) a second hydride transfer from *N*-DMBI-H to [Pt(COD)H][−] generates [Pt(COD)H₂][−], and (ii) this species transfers an electron to PDI-C₆C₇, regenerating the active catalyst Pt(COD)H₂ and completing the catalytic cycle. Importantly, in-situ absorption spectra acquired by exposing the n-doped PDI-C₆C₇ films (and doped PO12 as a polymer representative, Supplementary Fig. S20) to ambient air demonstrate identical absorption profiles with those of the pristine films, indicative of a complete recovery of the neutral semiconductors without any appreciable by-products (Fig. 4b). Similarly, quantitative ESR analysis of the catalytically n-doped PDI-C₆C₇ films revealed a strong ESR signal indicative of radical anions, which disappeared upon air exposure (Supplementary Fig. S21). This finding confirms that the charged species are indeed radical anions rather than hydride addition or decomposition products.

DFT calculations were performed to elucidate the role of Pt(COD)Cl₂ in enhancing the doping efficiency of OSCs. Specifically, the Gibbs free-energy profiles of PDI doping reactions by *N*-DMBI-H were compared in the presence and absence of the catalyst (Fig. 4e). Without catalyst, the most feasible reaction pathway has been established to follow a three-step process: (i) The C–H bond heterolytic cleavage enables hydride transfer from *N*-DMBI-H to PDI, generating *N*-DMBI⁺ and PDI-H[−], which requires $28.2 \text{ kcal mol}^{-1}$ and acts as the rate-determining step; (ii) PDI-H[−] then donates an electron to another PDI molecule, forming PDI-H[•]; and (iii) finally, two PDI-H[•] radicals combine to release H₂, regenerating PDI for the next cycle¹⁶. When Pt(COD)Cl₂ is introduced, the rate-limiting step remains the initial hydride transfer from *N*-DMBI-H. However, when the active catalyst Pt(COD)H₂ accepts the hydride instead of PDI-C₆C₇, the activation barrier is reduced by 9 kcal mol^{-1} . According to the Arrhenius equation, this reduction translates to a reaction rate increase of approximately 1×10^5 times at 120 °C, effectively demonstrating the catalyst's pivotal role in accelerating the doping reaction of OSCs.

Importantly, the mechanistic insights uncovered in this study offer clear guidelines for the rational design of the next-generation organometallic catalysts. Specifically, we propose that even higher n-doping efficiencies under milder thermal activation conditions could be achieved by tailoring both the electronic and steric environment surrounding the Pt(II) center. Since the activation mechanism involves cleavage of the Pt–Cl bond to enable hydride abstraction from the dopant, replacing the chloride ligands with more labile and less strongly coordinating anions represents a promising strategy to increase the Lewis acidity of the metal center and reduce the activation barrier^{39,40}. In parallel, modifying the COD ligand with alternatives that reduce steric hindrance or increased conformational flexibility may improve the catalyst's miscibility within polymer matrices⁴¹. This enhanced miscibility is expected to facilitate more uniform dispersion, promoting enhanced molecular ordering and higher crystallinity in doped films, both critical for achieving superior charge transport and device performance. Together, these structural modifications define a clear path toward the development of more efficient, low-temperature-compatible organometallic doping systems.

Application demonstration

The technological significance of this approach is demonstrated by the exceptionally high thermoelectric performance achieved by the PO12 polymer after Pt(COD)Cl₂-catalyzed doping with *N*-DMBI-H. As shown in Fig. 5a, the absolute Seebeck coefficient (*S*) decreases gradually with increasing catalyst loading, aligning with the expected reverse correlation between *S* and charge carrier concentration in the doping process (see Supplementary Fig. S24)⁴². However, at a comparable σ of 90 S cm^{-1} , we observed a 28% increase in *S* compared to PO12 films doped using the leading catalytic method currently available (from -88 to $-110 \text{ } \mu\text{V K}^{-1}$) (Fig. 5b)²³. To better understand this behavior, we examined the *S*– σ relationship for catalytically n-doped films prepared using either 5 mol% of Pt(COD)Cl₂ or pre-evaporated AuNPs as catalyst. As shown in Fig. 5c and Supplementary Fig. S25, Pt(COD)Cl₂-catalysed samples display a less pronounced decrease in *S* with increasing σ , thereby shifting the *S*– σ curve upward relative to the AuNP-catalysed films. Fitting these data to the power-law model proposed by Kang and Snyder, with a transport parameter of $s = 1$ as observed in comparable systems⁴³, reveals a higher transport coefficient for Pt(COD)Cl₂-catalysed films ($\sigma_0 = 23.50 \pm 1.17$) compared to those doped via the AuNP-catalysed approach ($\sigma_0 = 15.66 \pm 3.53$). These findings indicate that the improved σ observed with Pt(COD)Cl₂ is not merely due to increased carrier density, but also reflects enhanced charge carrier mobility^{44,45}. This enhancement is likely attributed to the catalyst's ability to effectively reduce the Coulombic interaction between the host and dopant, thereby mitigating structural disorder in heavily doped semiconductors (Fig. 5d)^{46–48}, as assessed here by comprehensive structural

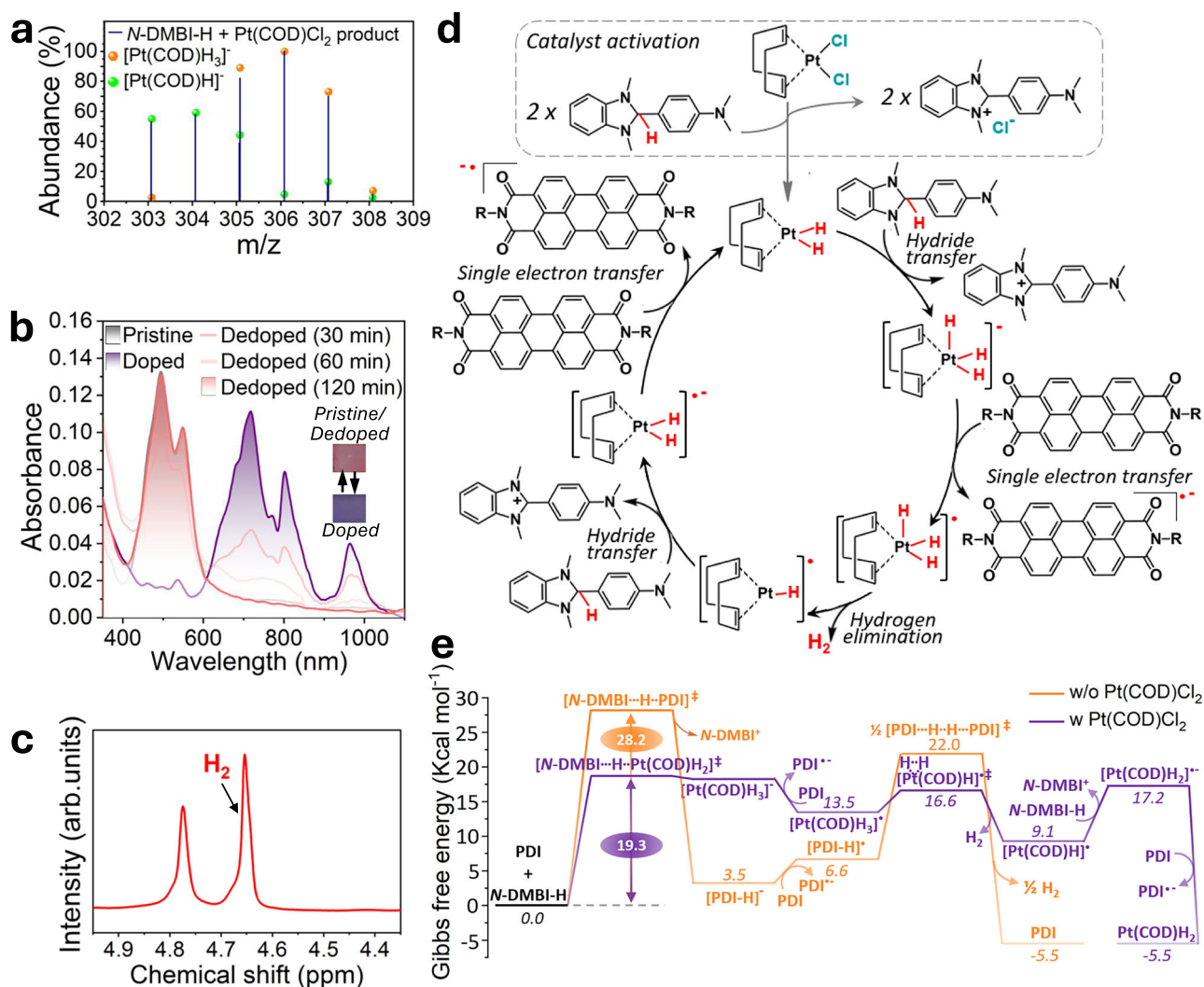


Fig. 4 | Mechanism of Pt(COD)Cl₂-catalysed n-doping by *N*-DMBI-H. **a** ESI-MS negative scan of 2 eq. *N*-DMBI-H + Pt(COD)Cl₂ solution (blue lines) as well as the predicted ESI-MS signals for [Pt(COD)H][−] (green dots) and [Pt(COD)H₃][−] (orange dots) anions. **b** UV-vis-NIR absorption spectra of pristine PDI-C₆C₇ film (gray line), *N*-DMBI-H doped films with 10 mol% Pt(COD)Cl₂ (purple line), and dedoped film after ambient air exposure (salmon line). The corresponding color changes for the doped/dedoped films are also shown. As exposure time increases, the absorption peaks of PDI-C₆C₇ regenerate from its radical anion, which was quenched by ambient air due to electron transfer reactions with O₂ and H₂O. Notably, dedoped films exhibit absorption features identical to their pristine forms. **c** ¹H-NMR spectrum measured during in-situ Pt(COD)Cl₂-catalysed *N*-DMBI-H doping of PDI-C₆C₇,

evidencing the H₂ signal at 4.6 ppm. ¹H-NMR experiments were performed in CD₂Cl₂, using sealed NMR tubes (298 K, 500 MHz). **d** Proposed catalytic cycle for the Pt(COD)Cl₂-catalysed *N*-DMBI-H doping of PDI-C₆C₇. **e** DFT-calculated Gibbs free-energy profile for the reaction with and without Pt(COD)Cl₂ catalyst. The calculations aimed at identifying the most plausible reaction mechanism were based on a comparative analysis of the energy profiles for various mechanisms consistent with the experimental data (see Supplementary Fig. S22 and 23). PDI-C₆C₇ was modelled by replacing 1-hexylheptyl groups with methyl groups. All calculations were performed at the M06/6-311 + G** level for C, H, N, and O atoms, and at the M06/LANL2DZ level for Pt.

analysis. Analyses by time-of-flight secondary ion mass spectrometry (ToF-SIMS) and photoinduced force microscopy (PiFM) reveal a homogenous distribution of Pt(COD)Cl₂ throughout the polymer matrix, indicating good miscibility at nanoscale (Supplementary Fig. S27, 28). Transmission electron microscopy (TEM) and atomic force microscopy (AFM) further confirm that the characteristic fibrillar nanostructure of the pristine PO12 film is preserved even at 10 mol % Pt(COD)Cl₂ loading, with no evidence of macrophase separation or formation of pure catalyst domains (Supplementary Fig. S29, 30). This observation is further supported by grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements, which demonstrate that catalytic doping with 5 mol% Pt(COD)Cl₂ effectively mitigates the microstructural disruptions induced under high doping conditions (5 mg mL^{−1} of *N*-DMBI-H). Specifically, the crystalline coherence lengths (CCLs) along the lamellar (100) and π - π stacking (010)

directions decrease from 87.0/36.9 Å in the pristine film to 68.6/29.6 Å in conventionally doped PO12, clearly indicating a pronounced loss of molecular ordering (Supplementary Fig. S32 and Table S9). Remarkably, catalytic doping restores the CCL₍₁₀₀₎/CCL₍₀₁₀₎ values to 80.8/36.2 Å, closely approaching those of the pristine film. Additionally, incorporation of Pt(COD)Cl₂ induces a slight increase in lamellar spacing (~1.5–1.8 Å), while the π - π stacking distance remains unchanged, suggesting that Pt(COD)Cl₂ preferentially localizes between the polymeric side chains. This spatial arrangement likely modulates polymer-dopant counterion interactions while minimizing structural perturbations along the conjugated backbone, thus preserving crystallinity under high doping levels. Additional evidence from X-ray photoelectron spectroscopy (XPS) and Raman spectroscopy, supported by DFT calculations, further corroborates this interpretation. XPS analysis shows that Pt(COD)Cl₂ significantly suppresses the peak shifts and

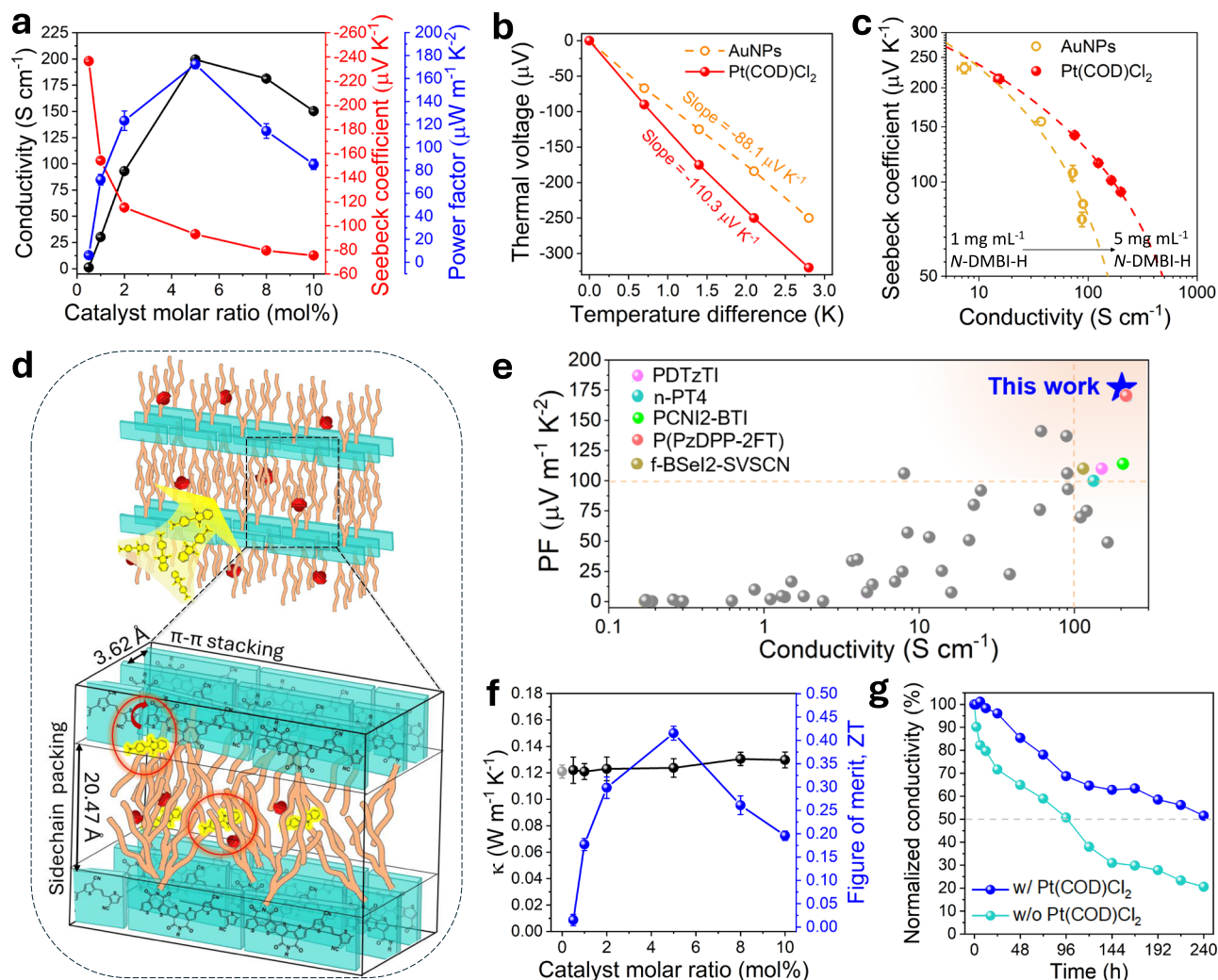


Fig. 5 | Thermoelectric properties of catalytically n-doped PO12 polymer with *N*-DMBI-H. **a** Electrical conductivity (black line), Seebeck coefficient (S , red line), and power factor (PF , blue line) of PO12 films doped with *N*-DMBI-H at varying loadings of Pt(COD)Cl₂ catalyst. Error bars indicate the standard deviation of five independent samples. **b** Linear fitting of the thermal voltage generated across catalytically doped PO12 films with Pt(COD)Cl₂ versus AuNPs at varying temperature differences. The results show a steeper slope for the linear curve of Pt(COD)Cl₂ catalysed n-doping, which translates into an improved S value. **c** σ - S relationship for polymer films catalytically n-doped in the presence of either 5 mol% of Pt(COD)Cl₂ or pre-evaporated AuNPs as catalyst, fitted using the power-law model proposed by Kang and Snyder. Doping was performed with increasing concentrations of *N*-DMBI-H (1–5 mg mL⁻¹, in 1 mg mL⁻¹ increments). **d** Schematic illustration of the influence of Pt(COD)Cl₂ on polymer-dopant counterion interactions. The presence

of Pt(COD)Cl₂ modulates polymer-dopant counterion interactions, leading to a reduction in counterion-induced structural disorder within the polymer matrix, which is crucial for improving the thermoelectric properties of the material. **e** Comparison of σ and PF values for catalytically *N*-DMBI-H doped PO12 with those of high-performance n-type OTE polymers reported in the literature (see Supplementary Table S8). OSCs with both σ and PF exceeding a value of 100 are highlighted. **f** Thermal conductivity (black line) and thermoelectric figure of merit ZT (blue line) of n-doped PO12 films as a function of Pt(COD)Cl₂ loading. The gray point represents the thermal conductivity measured in the absence of catalyst, corresponding to a conventional n-doping approach. **g** Temporal evolution of σ in n-doped PO12 films with and without 5 mol% Pt(COD)Cl₂ under inert-conditions. A dopant concentration of 5 mg mL⁻¹ was used.

broadening of the $S\ 2p$ signal of PO12 induced by conventional doping, consistent with a less disruptive doping environment (Supplementary Fig. S33)⁴⁹. Similarly, Raman spectra unveil that conventionally doped PO12 films exhibit upshifts of up to 4 cm⁻¹ and increased FWHM values of up to 7 cm⁻¹ in bands associated with C–C stretching modes prone to rotational disorder, indicative of strong electrostatic interactions between ionized *N*-DMBI⁺ species and the polymer backbone (Supplementary Fig. S34)⁵⁰. In contrast, catalytically doped films show substantially mitigated perturbations, with minimal broadening ($\sim 2\ cm^{-1}$) and no apparent peak shifts, reinforcing the conclusion that Pt(COD)Cl₂ enables a more controlled and less structurally disruptive doping process. These observations align with previous studies demonstrating that modulation of short-range polymer-dopant interactions can significantly improve charge transport^{51–53}.

As a result of the simultaneous enhancement in both σ and S , an exceptionally high power factor (PF , calculated as $PF = \sigma \cdot S^2$) of $175.4\ \mu W\ m^{-1}\ K^{-2}$ was achieved at a Pt(COD)Cl₂ concentration of 5 mol% (Fig. 5d). However, for practical thermoelectric applications, maintaining low thermal conductivity (κ) under optimized catalytic n-doping conditions is equally critical. To assess this, we measured the κ of doped films at room temperature ($T = 298\ K$) using the transient hot-wire method⁵⁴, enabling full determination of the dimensionless figure of merit ($ZT = \sigma S^2 T / \kappa$). Our measurements revealed that the incorporation of Pt(COD)Cl₂ exerts a negligible effect on κ of the doped systems. Specifically, conventionally n-doped PO12 films exhibited a κ of $0.121 \pm 0.005\ W\ m^{-1}\ K^{-1}$, lower than that of the pristine PO12 films ($0.142 \pm 0.011\ W\ m^{-1}\ K^{-1}$) due to enhanced phonon scattering caused by free carriers and dopant-induced microstructural

disorder^{55,56}. Importantly, κ remains nearly unchanged ($0.129 \pm 0.006 \text{ W m}^{-1} \text{ K}^{-1}$) even at the highest $\text{Pt}(\text{COD})\text{Cl}_2$ loading of 10 mol%, preserving the intrinsically low κ essential for thermoelectric applications. Consequently, at the optimal loading of 5 mol% $\text{Pt}(\text{COD})\text{Cl}_2$, the combination of excellent thermal insulation ($\kappa = 0.122 \text{ W m}^{-1} \text{ K}^{-1}$) and the outstanding PF yields an impressive ZT of 0.43 at 298 K, placing this result among the highest-performing n-type organic thermoelectrics reported to date^{8,57–61}. Notably, the power of our current doping approach lies not only in achieving impressive doping efficiency and enhanced PF without compromising the low κ required for efficient thermoelectric operation, but also in its ability to improve the stability of the doped films compared to those prepared via conventional n-doping (Fig. 5g and Supplementary Fig. S26). The stability of PO12 is intrinsically poor, primarily due to its insufficiently stabilized LUMO level (-3.81 eV) and the detrimental effects of oligo(ethylene glycol) (OEG)-type side chains, which promote moisture-induced electron trapping and broaden the interfacial charge transport density of states^{62,63}. As a result, the σ of conventionally n-doped PO12 rapidly decays to zero within minutes of air exposure and decreases to ~50% of its initial value after only 4 days under inert conditions²³, with long-term σ s expected to approach negligible values in line with observations for other n-type and even p-type OSCs^{64,65}. In sharp contrast, catalytically n-doped PO12 films not only demonstrate improved air stability (Supplementary Fig. S26) but also preserve nearly 50% of their initial σ even after 10 days in an inert atmosphere (Fig. 5g), highlighting the clear stability advantage of our catalytic strategy. These features set our strategy apart from previous catalytic n-doping methods, which generally lead to a trade-off of device performance^{22,49}.

In summary, we have developed a simple, solution-processable strategy for catalytic n-doping of OSCs that operates in a few seconds under mild activation conditions, offering on-demand control over doping levels. This strategy has been well validated for its broad applicability across various OSCs, dopants, and catalysts, consistently achieving high doping efficiencies and σ enhancements. In-depth characterizations together with DFT calculations well elucidate the reaction mechanism with the key intermediates identified through spectroscopic analysis. Additionally, this approach concurrently enhances electrical conductivity and mitigates microstructure disruption in films at high dopant loadings, an outcome that is otherwise challenging to attain with conventional or existing catalytically-assisted doping methods. The effectiveness of our strategy is demonstrated by an impressive σ over 200 S cm^{-1} and thermoelectric PF of $175 \mu\text{W m}^{-1} \text{ K}^{-2}$ achieved for doped PO12. Combined with excellent thermal insulation ($\kappa = 0.122 \text{ W m}^{-1} \text{ K}^{-1}$), this yields an impressive ZT of 0.43 at 298 K, placing it among the highest-performing n-type organic thermoelectrics reported to date. This approach thus opens a new horizon to explore organometallic complexes with enhanced catalytic activity and optimized counterion-induced effects in heavily doped semiconductors, establishing a powerful tool for advancing organic electronics.

Methods

Materials

The benchmark polymer N2200 and the $\text{PDI-C}_6\text{C}_7$ small molecule were obtained from Flexterra Corp. All solvents, as well as the *N*-DMBI-H, TAM dopants, and organometallic complexes, were purchased from Tokyo Chemical Industry (TCI) Co., Ltd., and were used as received without further purification unless otherwise specified. The imide-based polymers (PO12, f-BTI2g-SVSCN, and PDTzSI-Se), the cyano-functionalized polythiophene derivative (CNg4T2-CNT2), and the diketopyrrolopyrrole (DPP)-based polymer (PDCNBSe-DPP) were synthesized following previously reported procedures.

Synthesis of polymer PO12²³. A 5 mL vial was charged with monomer f-BTI2g-2Br (0.11 g, 0.10 mmol), 2,5-bis(trimethylstannyl)

thiophene (TVTCN-2Sn, 0.058 g, 0.10 mmol), tris(dibenzylideneacetone) dipalladium(0) ($\text{Pd}_2(\text{dba})_3$, 3.9 mg), and tris(*o*-tolyl)phosphine ($\text{P}(\text{o-tolyl})_3$, 10.0 mg). The reaction vessel was subjected to three pump–purge cycles with argon, after which anhydrous toluene (3 mL) was added. The vial was sealed under argon and heated at 140°C under microwave irradiation for 3 h. Upon cooling to room temperature, the reaction mixture was slowly dripped into 100 mL methanol. The resulting precipitate was collected by filtration and purified by sequential Soxhlet extraction with methanol, acetone, hexane, dichloromethane, and chloroform. The final chloroform fraction was concentrated and reprecipitated into methanol, affording polymer PO12 as a black solid (yield: 81%). ^1H NMR spectroscopy (400 MHz, CDCl_3 , δ in ppm): 7.1–7.0 (br, 6H), 4.4–4.3 (br, 4H), 3.5–3.3 (br, 40H), 3.3–3.1 (br, 12H), 2.4 (br, 2H). Molecular weight: $M_n = 23.8 \text{ kDa}$; $M_w = 46.2 \text{ kDa}$; $\bar{D} = 1.90$.

Synthesis of polymer f-BTI2g-SVSCN²⁵. A 5 mL vial was charged with f-BTI2g-2Br (62 mg, 0.052 mmol), 2,5-bis(trimethylstannyl)sele-nophene (SVSCN-2Sn, 34 mg, 0.052 mmol), $\text{Pd}_2(\text{dba})_3$ (0.7 mg), and $\text{P}(\text{o-tolyl})_3$ (1.9 mg). The vial was subjected to three pump–purge cycles with argon, after which anhydrous toluene (2.5 mL) was added. The reaction vessel was sealed under argon and heated at 140°C under microwave irradiation with stirring for 4 h. After cooling to room temperature, the mixture was precipitated into methanol (100 mL). The resulting solid was collected by filtration and purified by sequential Soxhlet extraction with methanol, acetone, hexane, dichloromethane, chloroform, and hexafluoroisopropanol. The final hexafluoroisopropanol fraction was concentrated and reprecipitated into methanol. The precipitate was collected and dried under vacuum to afford f-BTI2g-SVSCN as a black solid (yield: 68%). ^1H NMR spectroscopy (400 MHz, CDCl_3 , δ in ppm): 8.1–8.0 (br, 2H), 7.0–7.01 (br, 2H), 6.8–6.7 (br, 2H), 4.4–4.5 (br, 4H), 3.6–3.3 (br, 52H), 2.6–2.4 (br, 2H). Molecular weight: $M_n = 202.6 \text{ kDa}$; $M_w = 796.8 \text{ kDa}$; $\bar{D} = 3.93$.

Synthesis of polymer PDTzSI-Se²⁶. A 5 mL reaction tube equipped with a magnetic stir bar was charged with DTzSI-2Br (86.2 mg, 0.066 mmol), TVTCN-2Sn (37.5 mg, 0.066 mmol), $\text{Pd}_2(\text{dba})_3$ (0.90 mg, 0.001 mmol), $\text{P}(\text{o-tolyl})_3$ (2.41 mg, 0.008 mmol), and anhydrous toluene (3 mL). The tube was purged with argon, sealed under argon, and placed in a microwave reactor (Biotage Sweden AB, 400 W). The mixture was heated to 120°C and stirred for 15 min. After cooling to room temperature, the reaction mixture was dripped into 30 mL methanol, and the resulting precipitate was collected by filtration. The crude polymer was purified by sequential Soxhlet extraction with methanol, acetone, hexane, dichloromethane, and chloroform. The final chloroform fraction was re-dissolved in chloroform and reprecipitated into methanol, and the precipitate collected to afford polymer PDTzSI-Se as a dark solid (yield: 88%). ^1H NMR spectroscopy (400 MHz, $\text{C}_2\text{D}_2\text{Cl}_4$, δ in ppm): 8.2–7.9 (br, 2H), 7.4–6.9 (br, 4H), 4.9–4.0 (br, 4H), 4.0–3.1 (br, 52H), 2.6–2.4 (br, 2H). Molecular weight: $M_n = 145.8 \text{ kDa}$; $M_w = 319.1 \text{ kDa}$; $\bar{D} = 2.19$.

Synthesis of polymer CNg4T2-CNT2²⁷. A flame-dried glass tube was charged with CNg4T2-2Br (65 mg, 0.083 mmol), CNT2-2Sn (45 mg, 0.083 mmol), $\text{Pd}_2(\text{dba})_3$ (1.14 mg), and $\text{P}(\text{o-tolyl})_3$ (3.03 mg). The tube was subjected to three pump–purge cycles with argon, then anhydrous toluene (3 mL) was added. The reaction tube was sealed under argon and heated at 120°C for 60 min. After cooling to room temperature, the mixture was precipitated into methanol (100 mL) under vigorous stirring. The resulting solid was collected by filtration and purified by Soxhlet extraction using methanol, acetone, hexane, dichloromethane, and chloroform. The final chloroform fraction was concentrated, reprecipitated into methanol, and the precipitate collected and dried under vacuum to afford CNg4T2-CNT2 as a dark solid (yield: 68%). ^1H NMR spectroscopy (400 MHz, CDCl_3 , δ in ppm): 8.0–7.8 (br, 2H), 4.7–4.6 (br, 4H), 3.8–3.3 (br, 28H), 1.3–1.2 (br, 6H). Molecular weight: $M_n = 85.3 \text{ kDa}$; $M_w = 145.6 \text{ kDa}$; $\bar{D} = 1.71$.

Synthesis of polymer PDCNBSe-DPP²⁸. DCNBSe-2Br (43 mg, 0.11 mmol) and DPP-2Sn (130 mg, 0.11 mmol) were dissolved in degassed toluene (3 mL) in a reaction tube under argon. Tris(dibenzylideneacetone)dipalladium(0) ($\text{Pd}_2(\text{dba})_3$, 1.9 mg, 2 mol%) and tris(*o*-tolyl)phosphine ($\text{P}(\text{o-tolyl})_3$, 2.6 mg, 8 mol%) were quickly added under an argon atmosphere. The tube was sealed and stirred at 80 °C for 10 min, 100 °C for 10 min, and 140 °C for 3 h under microwave irradiation. Subsequently, 2-(tributylstannyl)thiophene (0.1 mL) and 2-bromothiophene (0.2 mL) were added, stirring at 110 °C for 30 min after each addition. The reaction mixture was then cooled to room temperature and precipitated into methanol (200 mL). The resulting crude polymer was collected by filtration and purified by Soxhlet extraction with methanol, acetone, and hexane. The product polymer was then dissolved in chloroform, reprecipitated into methanol, and collected by filtration. After drying under vacuum, PDCNBSe-DPP was obtained as a red solid (yield: 70%). ¹H NMR spectroscopy (400 MHz, CDCl_3 , δ in ppm): 9.0–8.9 (br, 2H), 8.6–8.4 (br, 2H), 4.3–3.9 (br, 4H), 2.1–1.9 (br, 2H), 1.3–1.1 (br, 64H), 0.9–0.7 (br, 12H). Molecular weight: $M_n = 69.4$ kDa; $M_w = 240.0$ kDa; $\bar{D} = 3.40$.

Sample preparation

Borosilicate glass (1.1 nm thick) or polyethylene terephthalate (PET, 125 μm thick) substrates were sequentially sonicated in isopropanol, acetone, and again in isopropanol for 25 min each. Afterward, they were dried using a stream of N_2 and treated with UV-ozone for 15 min. Thin films of the studied organic semiconductors were prepared under the following conditions (Summarized in Table S2): For PO12, 3 mg mL⁻¹ solutions in hexafluoroisopropanol (HFIP) were spin-coated at 1500 rpm for 60 s and annealed at 160 °C for 15 min, resulting in films with a thickness of ~35 nm; For f-BTI2g-SVSCN polymer, 5 mg mL⁻¹ solutions in HFIP were spin-coated at 1500 rpm for 60 s and annealed at 150 °C for 20 min, resulting in films with a thickness of ~31 nm; For PDTzSI-Se, 5 mg mL⁻¹ solutions in chloroform (CF) were spin-coated at 1500 rpm for 40 s and annealed at 90 °C for 15 min, resulting in films with a thickness of ~35 nm; For CNg4T2-CNT2 polymer, 5 mg mL⁻¹ solutions in HFIP were spin-coated at 1500 rpm for 60 s and annealed at 190 °C for 20 min, resulting in films with a thickness of ~63 nm; For PDCNBSe-DPP polymer, 4 mg mL⁻¹ solutions in *ortho*-dichlorobenzene (*o*-DCB) were spin-coated at 1000 rpm for 60 s and annealed at 200 °C for 15 min, resulting in films with a thickness of ~40 nm; For the benchmark N2200 polymer, 5 mg mL⁻¹ solutions in chlorobenzene (CB) were spin-coated at 1000 rpm for 60 s and annealed at 120 °C for 2 h, resulting in films with a thickness of ~25 nm; Finally, for PDI-C₆C₇ small molecule, 10 mg mL⁻¹ solutions in CF were spin-coated at 1000 rpm for 60 s and annealed at 120 °C for 30 min. In all cases, the samples were placed at a distance of ~3 cm from the center of the spin-coater (off-centre). Solutions of *N*-DMBI-H and TAM *n*-dopants were prepared at a concentration of 5 mg mL⁻¹ using *n*-butyl acetate as solvent. The COD-based organometallic complexes $\text{Pt}(\text{COD})\text{Cl}_2$, $\text{Pd}(\text{COD})\text{Cl}_2$, $[\text{Rh}(\text{COD})\text{Cl}]_2$, and $[\text{Ir}(\text{COD})\text{Cl}]_2$ were dissolved in acetonitrile at a concentration of 5 mg mL⁻¹. All solution preparation, film deposition, and thermal annealing steps were carried out in a N_2 -filled glovebox.

Catalytic *n*-doping of films

Sequential doping. The organic semiconductor and *n*-dopant solutions were spin-coated separately, resulting in two different procedures based on how the catalyst is introduced into the film: (a) For Seq-D doping, an organic semiconductor + catalyst blend solution was firstly spin-coated following the previously described conditions. Various molar ratios of catalyst relative to the organic semiconductor were used, ranging from 0.5 to 10% (where 10 mol% corresponds to 0.1 molar equivalent of its repeating unit). The spin-coated samples were then annealed at the desired temperature for a given time to enhance

their supramolecular order. Subsequently, a second solution containing 5 mg mL⁻¹ of dopant in *n*-butyl acetate was dropped onto the films, allowed to stand for 10 s, and finally spin-coated at 1500 rpm for 30 s to remove excess solution. (b) For Seq-D + C doping, the organic semiconductor was initially spin-coated and thermally annealed. After semiconductor film deposition, a second dopant + catalyst blend solution was spin-coated at 1500 rpm for 30 s onto the films immediately after preparation. Since the reaction starts when both catalyst and dopant become in contact, rapid mixing and casting of the dopant + catalyst blend solution are crucial to ensure accurate measurement of doping efficiency. A dopant concentration of 5 mg mL⁻¹ in *n*-butyl acetate was used in all cases, with catalyst molar ratios ranging from 0.5 to 10% (where 10 mol% corresponds to 0.1 molar equivalent of the dopant).

Blend doping. A ternary blend solution consisting of the organic semiconductor, the molecular *n*-dopant, and the organometallic catalyst was directly spin-coated at 1500 rpm for 60 s. A dopant concentration of 80 mol% (with 100 mol% representing one molar equivalent per repeating unit of the polymer) was used to enable direct comparison with previous studies²³. Additionally, small aliquots of catalyst solutions were then added to the semiconductor-dopant mixture to achieve the desired catalyst-to-dopant molar ratio, typically in the range of 0.5–10 mol%.

In the presence of organometallic complexes, both sequential and doping procedures were completed with a mild thermal annealing of 120 °C for 10 s to activate the doping, while annealing times of 10 min (or 1 h for N2200) were required in the absence of catalyst. The optimal doping conditions yielding the highest electrical conductivity for the studied organic semiconductors are summarized in Table S2.

Electrical measurements

30 nm Au electrodes with 100 μm channel length (L) and 2000 μm channel width (W) were thermally evaporated through a shadow mask. For devices with AuNP catalyst, a nominal thickness of 0.1 nm Au was thermally evaporated prior to electrode deposition. The linear I - V curves were characterized in a N_2 -filled glove box using a Keithley 4200-SCS semiconductor characterization system. Electrical conductivities were then calculated using the standard equation $\sigma = (1/R) \times [L/(W \times h)]$, where R is the device resistance and h is the film thickness measured with a Tencor KLA D-120 profilometer. Average electrical conductivity values were calculated from five different measurements.

UV-vis-NIR absorption

Thin films were prepared on borosilicate glass substrates and subsequently loaded into tightly sealed glass cuvettes within a N_2 -filled glovebox to prevent exposure to ambient atmosphere. The optical absorption spectra were recorded using a Shimadzu UV-3600 UV/vis/NIR spectrophotometer with a resolution of 2 nm.

ESR

The thin films were prepared on PET substrates, which were then cut into 3×12 mm pieces and loaded into tightly sealed quartz ESR tubes inside a N_2 -filled glovebox. Measurements were performed on an X-band (9.84 GHz) Bruker EMXplus-10 spectrometer, using a magnetic field modulation magnitude of 2 gauss, a microwave power of 2 mW, and a modulation frequency of 100 kHz. Spin-counting was carried out using Bruker software following standard procedures.

¹H-NMR

¹H NMR spectra were recorded on a Bruker Ultra Shield Plus AV400 spectrometer. Samples were prepared inside a N_2 -filled glovebox using $\text{CDCl}_3/\text{CD}_2\text{Cl}_2$ as solvents.

ESI-MS

The sample was prepared under N₂ atmosphere by dissolving *N*-DMBI-H and Pt(COD)Cl₂ in dry CH₂Cl₂ at a 2:1 molar ratio. The reaction solution was further diluted to a concentration below 0.1 mg mL⁻¹ before data collection using a Thermo Scientific LCQ Fleet system under ESI mode.

AFM

Thin films were prepared on borosilicate glass and characterized in ambient air using a Dimension Icon Scanning Probe Microscope (Asylum Research, MFP-3D-Stand Alone) working in tapping mode.

TEM

Samples were prepared by spin-coating the films onto 40 nm PEDOT:PSS-coated borosilicate glass substrates, following the optimal device fabrication conditions. After drying, the substrates were floated on deionized water, and the released films were carefully transferred onto copper TEM grids. The samples were then placed in a vacuum desiccator overnight to ensure complete drying. TEM images were acquired using a Tecnai G2 F30 transmission electron microscope operating at 300 kV.

ToF-SIMS

Blend PO12-Pt(COD)Cl₂ films were prepared on Si/SiO₂ substrates and analysed under ultra-high vacuum conditions (~10⁻⁹ mbar) using an ION TOF SIMS 5 system equipped with a Bi⁺ primary ion source operated at 30 keV. A time-of-flight mass analyser provided high mass resolution, achieving a full width at half maximum (FWHM) of >11,000 for low-mass ions (~29 u) and >16,000 for high-mass ions (>200 u). Dual sputtering ion guns (O₂ and Cs⁺) were employed for depth profiling, operated at energies from 250 eV to 2 keV and enabling both positive and negative ion detection. The system achieved a lateral spatial resolution better than 100 nm and a depth resolution of approximately 2 nm.

PiFM

Blend PO12-Pt(COD)Cl₂ films were prepared on Si/SiO₂ substrates and characterized using a VistaScope microscope (Molecular Vista, Inc.). PiFM measurements were performed in sideband excitation mode, with the laser frequency modulated at $f_m = f_1 - f_0$, where f_0 is the first mechanical eigenmode resonance of the cantilever used for PiFM signal detection, and f_1 is the second eigenmode, simultaneously employed for acquiring the AFM topography. A Pt-coated tip (PPP-NCHPt-MB, Nanosensors) with a resonant frequency of ~350 kHz was used throughout the experiments.

GIWAXS

OSC thin films were prepared on Si/SiO₂ substrates and measured at the Beamline PLS-II 9A SAXS of Pohang Accelerator Laboratory, Republic of Korea. The X-ray energy was 11.09 eV and the incidence angle was 0.12°. The samples were measured in vacuum and the total exposure time was 10 s. The scattered X-rays were recorded by a CCD detector located 222.119 mm from the sample. All samples for GIWAXS measurements had a similar thickness of around 40 nm.

XPS

X-ray photoelectron spectroscopy measurements were performed using a Thermo ESCALAB 250Xi instrument equipped with a monochromatic Al K α (h ν = 1486.6 eV) X-ray source at 150 W, and an X-ray spot size of 500 μ m. Thin films were prepared in a N₂-filled glovebox and subsequently sealed in a vacuum transfer device to maintain integrity. Measurements were performed under high-vacuum conditions (<10⁻⁸ mBar) throughout the testing process.

Raman

Raman spectra with 532 nm laser excitations were recorded for thin film samples by using a HORIBA LabRAM HR Evolution laser microscopy confocal Raman spectrometer equipped with an 800 mm focal length monochromator, a confocal microscope with a $\times 100$ objective and a high-performance CCD detector. This set up has a spectral resolution lower than 1 cm⁻¹.

DFT calculations

Geometry optimizations of all reactants, products, intermediates, and transition states were carried out along the entire catalytic cycle. Calculations were performed adopting the M06 hybrid meta-GGA functional. The effective core potential of Hay and Wadt (LANL2DZ) and the relative basis set were used for the platinum atoms. The standard all-electron 6-311 + G** basis set was used for all remaining atoms. Molecular geometry optimization of stationary points was carried out without symmetry constraints and used analytical gradient techniques. The transition states were searched with the “distinguished reaction coordinate procedure” along the emerging bonds. Frequency analysis was performed to obtain thermochemical information about the reaction pathways at 298 K upon the harmonic approximation. Solvation contribution to the energy profile were evaluated within the SMD approach only for molecular species using dichloromethane (ϵ = 8.93) as prototypical solvent. All calculations were performed using G16 code on Linux cluster systems.

Thermoelectric devices

Doped films of PO12 were prepared on borosilicate glass substrates under the same conditions as those used for electrical conductivity measurements. In this case, the substrates had a channel length of 1 mm, channel width of 12 mm, and electrode width of 500 μ m. A pair of Peltier devices with a separation distance of 1 mm were used to generate a temperature difference (ΔT) across the samples, monitored by a thermocouple. ΔT was systematically varied from 0 to 2.8 K in steps of 0.7 K, and the corresponding thermal voltages were recorded using a Keithley 4200-SCS semiconductor characterization system located in a N₂-filled glove box. Seebeck coefficients were extracted by linear fitting of the thermal voltage versus ΔT . Thermal conductivity (κ) was measured using a XIATECH TC3000E instrument, which operates based on the transient hot wire (THW) method, an established technique known for its high accuracy and precision in measuring the thermal conductivity of thin films.

Data availability

Data supporting the findings of this study are available in the Supplementary Information/Source Data file. The crystallographic data for the structure of *N*-DMBI⁺Cl⁻ have been deposited at the Cambridge Crystallographic Data Centre under deposition number CCDC 2449210. Source data are provided with this paper. All data are available from the corresponding author upon request. Source data are provided with this paper.

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Acknowledgements

X.G. is thankful for the financial support from the National Natural Science Foundation of China (52173171) and the Shenzhen Science and Technology Innovation Commission (JCYJ20220818100617037 and KCXST2022102111413031). K.F. acknowledges the financial support by the National Natural Science Foundation of China (22275078). A. Motta acknowledges CINECA award no. HP10CL5217 2024 under the ISCRA initiative for computational resources. H.Y.W. acknowledges the financial support from the National Research Foundation of Korea (RS-2025-24683763). The authors would like to thank Dr. Jiawei Huang and Dr. Xiaoyue He from Songshan Lake Materials Laboratory for their

assistance in PiFM measurements. The authors also acknowledge the assistance of SUSTech Core Research Facilities.

Author contributions

S.G.V., J.L. and X.G. conceived the project. S.G.V. prepared the samples and performed the device performance measurements, n-doping analyses and spectroscopic characterizations, including UV–vis-NIR absorption and ESR characterization. K.F., Y.Li. and S.M. synthesized the tested organic semiconductors. S.G.V. and Hanqiang W. conducted the mechanism studies. A.M. performed the DFT calculations regarding the reaction mechanism. B.Liu performed the TEM analysis. X.Z. contributed to the determination of film thickness. B.Li assisted in the XPS and AFM measurements. S.G.V. and Y.Lu. carried out the thermal conductivity measurements under the supervision of Hong W. S.Y.J. and H.Y.W. performed the GIWAXS measurements. S.G.V. and X.G. wrote the paper. X.G., A.M. and Hanqiang W. supervised this work. All authors contributed to data analysis and discussions.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-025-66052-9>.

Correspondence and requests for materials should be addressed to Hanqiang Wang, Alessandro Motta or Xugang Guo.

Peer review information Nature Communications thanks Yunfeng Deng, Yang Wang, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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